

INDIVIDUAL PROJECT SUMMARY REPORT

Calibo AI Academy – Phase 1 Mini Use Case 01

Student Name	MADA SURESH
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Project Title	Category Intelligence – Retail Product Performance EDA
Individual Role	Insights Lead – Step 3 (Product-Level Analysis) and Step 4 (Discount Effectiveness)
Tools Used	Python, Pandas, Matplotlib, Jupyter Notebook

CBIM Problem Canvas

Business Problem: Retail sales data contains product-level and discount information, but raw transaction data does not directly show which products perform strongly within each category or how different discount levels affect revenue.

Business Objective: Analyze product-level revenue performance and discount effectiveness to convert transaction data into clear insights that can support supplier review and promotional decision-making.

Data Used: Retail Sales Dataset containing 107,836 transactions across five categories from January 2026 to June 2026.

Key Analysis Focus: Step 3 – Product-Level Analysis; Step 4 – Discount Effectiveness.

Expected Business Output: A clear comparison of top and bottom products within each category and an evidence-based understanding of how discount levels relate to average revenue per transaction and units sold.

1. Project Overview

The project focuses on analyzing retail sales data to understand product-level performance and the effectiveness of different discount levels. My contribution focuses on converting transaction data into product comparisons and measurable discount insights that can support the team's overall business analysis.

2. My Individual Contribution – Insights Lead

As the Insights Lead, I worked on Step 3 and Step 4 of the Category Intelligence use case. In Step 3, I grouped the data by category and product, calculated total six-month revenue, and identified the top two and bottom two products within each category. This helped highlight stronger and weaker products for further supplier review.

In Step 4, I analyzed discount effectiveness by comparing different discount brackets. I calculated average revenue per transaction and average units sold for each discount level and used the results to understand whether higher discounts were associated with improved sales performance.

3. Step 3 – Product-Level Analysis

Purpose: Identify the highest- and lowest-performing products within each retail category based on total six-month revenue.

Method: Group the data by category and product, aggregate total revenue, sort products within each category, and select the top 2 and bottom 2 products.

Key Findings: In Electronics, Headphones generated approximately ₹47.69 million in six-month revenue and Laptop generated approximately ₹46.55 million. Smart TV was among the lower-performing Electronics products at approximately ₹41.88 million.

Business Interpretation: Product-level comparison helps identify products that contribute strongly to category revenue and products that may require further supplier review or corrective action.

4. Step 4 – Discount Effectiveness

Purpose: Understand how different discount levels relate to average revenue per transaction and average units sold.

Method: Group transactions into the available discount levels and calculate average revenue per transaction and average units sold for each level.

Key Findings: Average revenue per transaction was approximately ₹4,685.65 at 0% discount, ₹4,303.73 at 5%, ₹4,005.55 at 10%, ₹4,069.16 at 15%, and ₹3,631.27 at 20%. Average units sold remained relatively stable across these

discount levels.

Business Interpretation: In this dataset, higher discounts were not associated with higher average revenue per transaction. Since average units sold remained relatively stable, discount increases should be evaluated carefully rather than assuming that larger discounts will increase sales volume.

5. Skills and Concepts Applied

Data grouping and aggregation; Product-level revenue analysis; Top and bottom product identification; Discount-bracket analysis; Average revenue and units-sold calculations; Exploratory Data Analysis (EDA); Business interpretation; Data visualization using Matplotlib.

6. Learning Outcomes

Through this role, I learned how to move from transaction-level data to product-level business insights. I improved my understanding of grouping and aggregation, ranking products within categories, analyzing discount patterns, interpreting numerical results, and communicating findings in a business-oriented manner.

7. Conclusion

My contribution as the Insights Lead covered product-level performance and discount effectiveness in the Category Intelligence use case. Step 3 helped identify stronger and weaker products within categories, while Step 4 provided evidence on how discount levels related to revenue per transaction and units sold. Together, these findings support the team's supplier-review and promotional decision discussions.